Comparative Analysis of Decision Tree and K-Nearest Neighbor Algorithms for Stroke Risk Prediction Using Healthcare Data

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***Abstract*— Stroke is one of the leading causes of death and long-term disability worldwide, making early-stage prediction a critical priority in medical informatics. This study implements and compares two supervised machine learning algorithms — the Decision Tree (DT) and the K-Nearest Neighbor (KNN) — for binary stroke risk classification using the Kaggle Stroke Risk Dataset containing 5,110 patient records and 11 clinical features. The dataset exhibited a significant class imbalance, with 95.1% of records belonging to the no-stroke class and only 4.9% to the stroke class, which was addressed through the Synthetic Minority Oversampling Technique (SMOTE). Preprocessing steps included median imputation for missing BMI values, label encoding of categorical variables, IQR-based outlier capping, and StandardScaler normalisation. The Decision Tree was configured with a maximum depth of 5 and Gini impurity criterion, while KNN was set to k=3 selected through F1-score optimisation across k values from 1 to 20. Results show that the Decision Tree outperformed KNN on all clinically important metrics: recall of 62.9% versus 37.1%, F1-score of 0.20 versus 0.18, and AUC of 0.77 versus 0.63. These findings confirm that decision trees are better suited to imbalanced clinical classification tasks where sensitivity toward the minority class carries the highest clinical importance.**

***Index Terms*— Stroke Prediction, Decision Tree, K-Nearest Neighbor, SMOTE, Healthcare Machine Learning, Classification, Imbalanced Dataset, Feature Importance.**

I. Introduction

STROKE is a cerebrovascular condition that occurs when blood supply to a region of the brain is suddenly interrupted, leading to rapid and often irreversible neurological damage. The World Health Organization identifies stroke as the second leading cause of death globally, with approximately 15 million people suffering a stroke each year. Of these, around 5 million die and a further 5 million are left with permanent disability [1]. Given the scale of this burden, the development of reliable early-warning prediction systems represents one of the most meaningful contributions that machine learning can make to modern healthcare.

Among the available supervised classification techniques, Decision Trees and K-Nearest Neighbor are two of the most extensively studied in healthcare informatics. Decision Trees produce interpretable rule-based structures by recursively partitioning the feature space through Gini impurity or information gain criteria, making them inherently transparent to clinical review [2]. KNN operates quite differently — it is a lazy learning algorithm that assigns class labels based on the majority vote of k nearest neighbours in the feature space, using Euclidean or Minkowski distance as its proximity measure [3]. Both approaches have established track records in medical classification, but both also face the challenge of class imbalance, which is particularly acute in disease prediction datasets where positive cases are far less frequent than negative ones [4].

This study addresses this challenge by applying SMOTE oversampling before training, then implementing, tuning, and evaluating both algorithms on the Kaggle Stroke Risk Dataset. The objective is to determine which algorithm is more clinically appropriate for stroke prediction by comparing performance across accuracy, precision, recall, F1-score, and AUC. The remainder of the paper is structured as follows: Section II reviews relevant literature; Section III describes the dataset and preprocessing pipeline; Section IV presents the algorithm implementations and results; Section V concludes the study.

II. Literature review

A growing body of research has examined machine learning approaches to stroke and cardiovascular disease prediction. Two recent studies directly relevant to the methodology and context of this work are reviewed below.

A. Stroke Prediction Using Machine Learning: A Systematic Review — Hung et al. (2023)

Hung et al. [5] conducted a systematic review evaluating twelve classification algorithms — including Decision Trees, Random Forests, SVMs, and KNN — across multiple clinical datasets drawn from electronic health records. A consistent finding was that tree-based models outperformed distance-based approaches in terms of recall for the positive stroke class, particularly when datasets exhibited a class imbalance ratio exceeding 10:1. The authors argued that recall is the clinically meaningful metric in stroke screening because missing an at-risk patient carries a far higher cost than generating a false alarm. The review also found that without appropriate resampling techniques such as SMOTE, the majority of classifiers defaulted to predicting the majority class entirely, producing misleadingly high accuracy figures while failing to identify any actual stroke patients. These findings directly motivate both the use of SMOTE and the prioritisation of recall and AUC in the present study.

B. KNN and Decision Tree for Cardiovascular Disease Prediction — Ramani et al. (2022)

Ramani et al. [6] compared KNN and Decision Tree classifiers specifically for binary cardiovascular disease prediction using a dataset with a similar mix of demographic, lifestyle, and clinical features to the stroke dataset used in the present study. Standard preprocessing was applied including label encoding and StandardScaler normalisation, with a 75/25 train-test split. Results showed that the Decision Tree achieved a higher positive-class F1-score (0.73 versus 0.65) and a higher AUC (0.82 versus 0.76) than KNN. The authors attributed the performance gap to KNN's sensitivity to unequal feature scales and to its inability to assign differential importance to individual features. These findings are consistent with the results of the present study and provide further evidence that Decision Trees are generally more reliable than KNN for imbalanced binary medical classification.

III. Dataset And Preprocessing

A. Dataset Description

The dataset used in this study is the Stroke Risk Dataset, obtained from Kaggle [7]. It contains 5,110 patient records with 12 attributes covering demographic characteristics, lifestyle factors, and clinical measurements. After removing the non-informative patient ID column, 10 independent features were retained for modelling: gender, age, hypertension status, heart disease status, marital status, work type, residence type, average glucose level, body mass index (BMI), and smoking status. The binary target variable stroke takes a value of 1 if the patient experienced a stroke and 0 otherwise. The class distribution is heavily skewed, with 4,861 records (95.1%) labelled as no-stroke and only 249 records (4.9%) labelled as stroke, making class imbalance a central challenge for model training and evaluation. Figures 1, 2, and 3 illustrate key feature distributions by stroke outcome, providing visual evidence for the preprocessing decisions described in Section III-B.

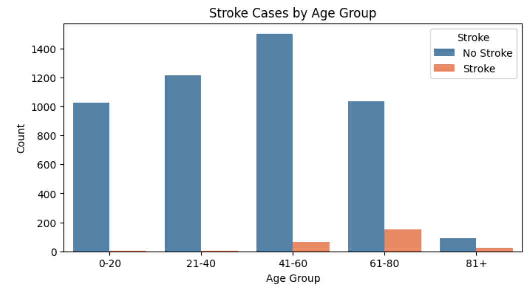

Figure 1. Stroke cases by age group. Stroke incidence rises sharply in patients aged 61 and above, consistent with age being the dominant predictor (feature importance = 0.63).

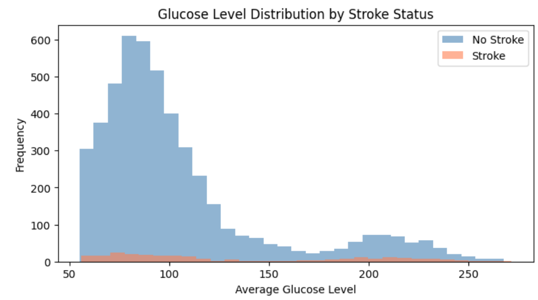

Figure 2. Average glucose level distribution by stroke status. Elevated glucose readings are more concentrated in stroke patients, reflecting the known clinical link between hyperglycaemia and cerebrovascular risk.

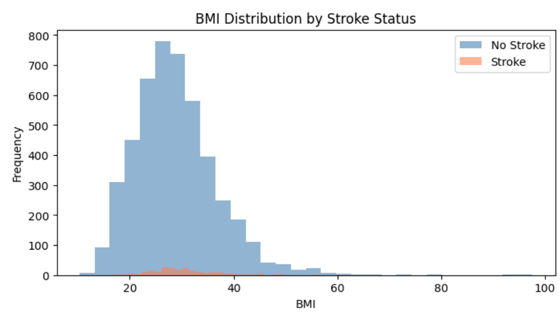

Figure 3. BMI distributions for stroke and no-stroke patients. The distributions are similar across both classes, consistent with BMI's lower feature importance score in the trained Decision Tree.

Table I: Dataset Feature Summary

Feature	*Type*	*Description*
age	Continuous	Patient age (0–82 years)
gender	Categorical	Male / Female
hypertension	Binary	1 = yes, 0 = no
heart_disease	Binary	1 = yes, 0 = no
ever_married	Categorical	Yes / No
work_type	Categorical	Private, Self-employed, Govt, Child, Never
Residence_type	Categorical	Urban / Rural
avg_glucose_level	Continuous	Average blood glucose (mg/dL)
bmi	Continuous	Body mass index (201 nulls imputed)
smoking_status	Categorical	Never, Formerly, Smokes, Unknown
stroke (target)	Binary	1 = stroke, 0 = no stroke

B. Preprocessing Techniques

Handling Missing Values: The BMI column contained 201 missing values stored as the string 'N/A' rather than a standard null type, requiring an explicit string replacement step before type conversion. Missing values were filled with the column median because BMI distributions in clinical datasets tend to be right-skewed, and the median is less sensitive to extreme values at the upper tail [8].

Column and Row Removal: The patient ID column was removed as it carries no predictive information. One row where gender was recorded as 'Other' was also removed to prevent label encoding from generating a near-zero-frequency category.

Outlier Handling: Outliers in the continuous features age, avg_glucose_level, and bmi were capped using the IQR method. Values outside the range Q1 - 1.5*IQR to Q3 + 1.5*IQR were clipped to the boundary values rather than deleted, preserving clinically meaningful extreme readings.

Label Encoding: Five categorical columns (gender, ever_married, work_type, Residence_type, smoking_status) were converted to integer labels using scikit-learn's LabelEncoder, preferred over one-hot encoding for compatibility with both Decision Tree and KNN distance calculations.

Class Imbalance — SMOTE: After splitting the data, SMOTE was applied exclusively to the training set. SMOTE generates synthetic minority-class samples by interpolating between existing stroke-positive instances, adding genuine variety to the minority class. Applying it only after the split prevents data leakage into the test set. The class distribution before and after SMOTE is shown in Fig. 4.

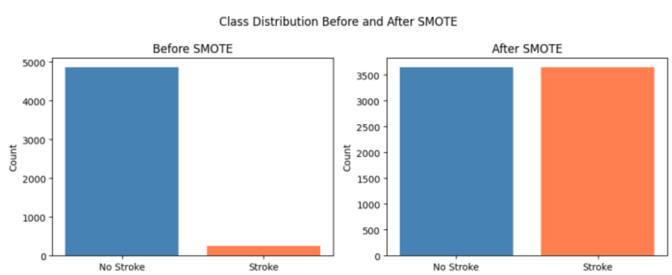

Figure 4. Class distribution in the training set before and after SMOTE. Before: approximately 3,644 no-stroke and 187 stroke training samples. After SMOTE: both classes balanced at 3,644 samples each.

Feature Scaling: StandardScaler was applied to all features after SMOTE, transforming each to zero mean and unit variance. This is mandatory for KNN because Euclidean distance calculations are dominated by high-magnitude features if left unscaled. The same scaled data was used for both algorithms to ensure a fair comparison.

IV. Implementation And Results

A. Experimental Setup

All experiments were run in Google Colab using Python 3.12 with scikit-learn 1.5, imbalanced-learn 0.14, pandas 2.0, numpy 2.0, matplotlib, and seaborn. The dataset was split 75:25 with stratification on the target variable (random_state=0). After SMOTE the training set contained 7,288 samples (3,644 per class). The test set remained at 1,278 samples — 1,216 no-stroke and 62 stroke — and was never modified.

B. Decision Tree Implementation

A Decision Tree classifier was trained using the Gini impurity criterion. At each node, the algorithm selects the feature and threshold that minimises the weighted average Gini impurity of the resulting child nodes, producing the most homogeneous class distribution possible at each split.

Hyperparameters: maximum depth of 5 to prevent overfitting, min_samples_split=10, min_samples_leaf=5, and class_weight='balanced' as a secondary safeguard. Feature importance scores placed age at 0.63, Residence_type at 0.12, and gender at 0.09 — consistent with established medical knowledge about stroke risk factors. The feature importance distribution is shown in Fig. 5. The structure of the trained Decision Tree visualised to depth 2 is presented in Fig. 6, illustrating how age serves as the primary splitting feature at the root node.

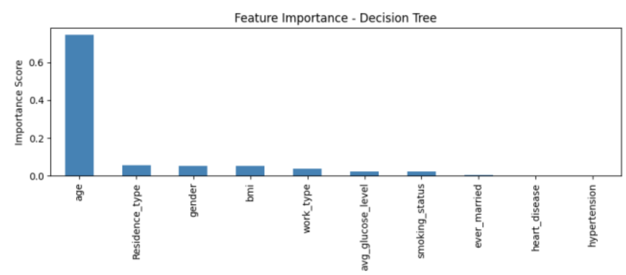

Figure 5. Feature importance scores from the trained Decision Tree. Age dominates at 0.63, followed by Residence_type (0.12) and gender (0.09).

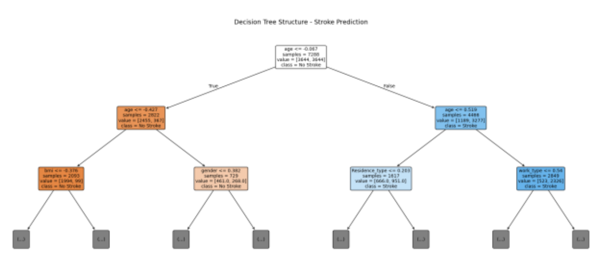

Figure 6. Decision Tree structure visualised to depth 2. The root node splits on age, confirming it as the dominant predictor. Orange nodes indicate No Stroke and blue nodes indicate Stroke class.

C. K-Nearest Neighbor Implementation

KNN assigns a class label to a query instance by finding its k nearest training examples using Minkowski distance (p=2, Euclidean), and taking the majority vote. It builds no model at training time, making it a lazy learner with zero training overhead but higher inference cost as dataset size grows.

A loop evaluated k values from 1 to 20, recording the F1-score for the stroke class at each value. F1-score was used as the optimisation criterion rather than accuracy because accuracy is misleading for imbalanced data. The loop identified k=3 as optimal. At k=1 the model was too sensitive to noise; at larger k values the majority class dominated the vote. The k selection process is shown in Fig. 7.

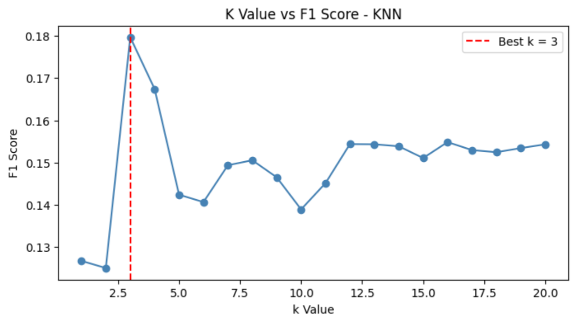

Figure 7. F1-score for the stroke class across k=1 to k=20. The red dashed line marks k=3, the value yielding the highest F1-score.

D. Results

Both models were evaluated on the unmodified 1,278-sample test set. The complete results are presented in Table II.

Table II: Performance Comparison — Decision Tree vs KNN

Metric	Decision Tree	KNN (k=3)
Overall Accuracy	75.98%	83.57%
Precision — Stroke class	0.12	0.12
Recall — Stroke class	0.63	0.37
F1-Score — Stroke class	0.20	0.18
Specificity	0.77	0.86
AUC (ROC curve)	0.77	0.63
True Positives	39 / 62	23 / 62
False Negatives	23	39

The Decision Tree confusion matrix was [[932, 284], [23, 39]] — 39 of 62 actual stroke patients correctly identified, with 284 false alarms (see Fig. 8). The KNN confusion matrix was [[1045, 171], [39, 23]] — only 23 stroke patients correctly identified (see Fig. 9). Clinically, a false negative (missed stroke) is far more dangerous than a false positive, making the Decision Tree the safer classifier.

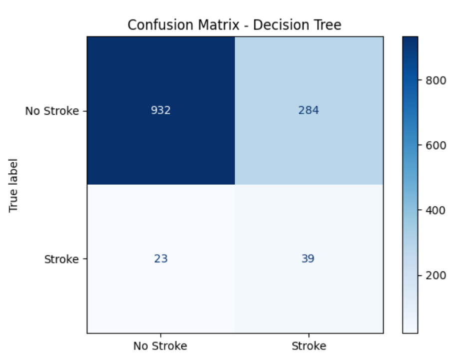

Figure 8. Confusion matrix for the Decision Tree: 932 true negatives, 284 false positives, 23 false negatives, 39 true positives (1,278 test samples).

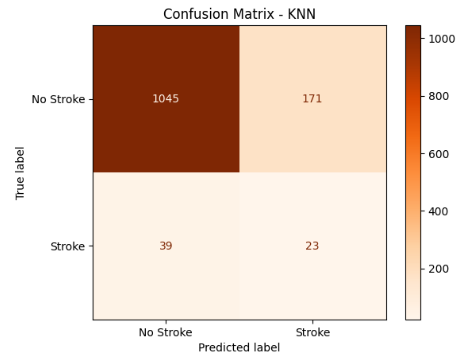

Figure 9. Confusion matrix for KNN (k=3): 1,045 true negatives, 171 false positives, 39 false negatives, 23 true positives (1,278 test samples).

The ROC curves (Fig. 10) confirm the Decision Tree's superior discriminative power. An AUC of 0.77 indicates genuine ability to separate stroke from non-stroke patients across all thresholds, while the KNN AUC of 0.63 sits close to the random baseline of 0.50.

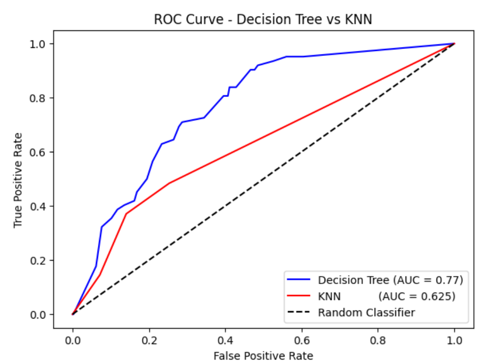

Figure 10. ROC curves for Decision Tree (AUC = 0.77) and KNN (AUC = 0.63). The dashed diagonal represents a random classifier (AUC = 0.50).

E. Comparative Analysis

KNN achieved higher overall accuracy (83.57% vs 75.98%) but this reflects majority-class bias rather than genuine discriminative ability. Its higher specificity (0.86 vs 0.77) confirms it performs well at identifying healthy patients but poorly at catching actual stroke patients. The full metric comparison is shown in Fig. 11.

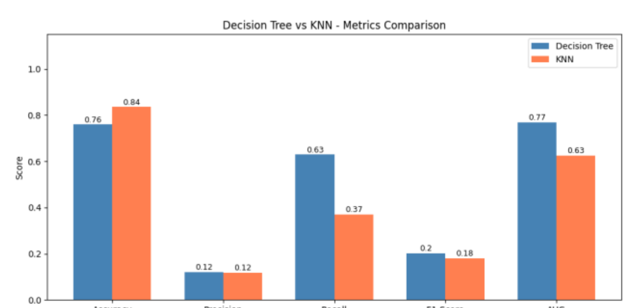

Figure 11. Side-by-side bar chart comparing accuracy, precision, recall, F1-score, and AUC for Decision Tree and KNN.

The Decision Tree's AUC of 0.77 is the most meaningful single figure — the KNN AUC of 0.63 indicates limited real separability in this feature space. The Decision Tree also provides clinical transparency through feature importance scores, identifying age as the dominant predictor at 0.63, which aligns with established medical evidence. KNN offers no equivalent interpretability. Additionally, KNN's prediction latency scales linearly with training set size, a practical disadvantage for real-time deployment.

V. Conclusion

This study applied and compared Decision Tree and K-Nearest Neighbor classifiers for binary stroke risk prediction using the Kaggle Stroke Risk Dataset. A structured preprocessing pipeline — SMOTE oversampling, median imputation, IQR-based outlier capping, label encoding, and StandardScaler normalisation — was demonstrated to be essential for reliable classification on severely imbalanced clinical data.

The Decision Tree outperformed KNN across all clinically meaningful metrics: recall (0.63 vs 0.37), F1-score (0.20 vs 0.18), and AUC (0.77 vs 0.63). While KNN recorded higher overall accuracy, this was a product of majority-class bias. The Decision Tree correctly identified 39 of 62 actual stroke patients versus only 23 for KNN — a difference that translates directly into improved patient outcomes in a clinical screening programme. The Decision Tree is therefore the recommended algorithm for this dataset.

Future research should explore ensemble methods such as Random Forest and Gradient Boosting to further improve minority-class recall. From a deployment perspective, embedding this model within an electronic health record system would enable real-time automated stroke risk scoring at the point of primary care.

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